

10 Milk and Dairy Products

10.1 Milk

Milk is the secreted fluid of the mammary glands of female mammals. It contains nearly all the nutrients necessary to sustain life. Since the earliest times, mankind has used the milk of goats, sheep and cows as food. Today the term “milk” is synonymous with cow’s milk. The milk of other animals is spelled out, e. g., sheep milk or goat milk, when supplied commercially.

In Germany, the yield of milk per cow in kg/year has increased steadily as a result of selective breeding and improvements in feed. The yield was 1260 kg per cow in 1812, 2163 kg in 1926, 3800 kg in the FRG in 1970, 4181 kg in 1977 and 6537 kg in 2003. In the EU in 2003, Swedish cows were the best performers at 8073 kg, followed by Danish and Dutch animals at 7889 kg and 7494 kg respectively.

In some countries it is permitted to increase the yield of milk by injection of the growth hormone bovine somatotropin (BST). The recombinant BST (rBST) used is identical in activity to natural BST. This is done by taking, from the DNA of cows, the specific gene sequence that carries the instructions for preparing BST and inserting it into *E. coli*, which can then produce large amounts of rBST. Natural BST consists of 190 or 191 amino acids. rBST may differ slightly in that a few extra amino acids may be attached at the N-terminal end of the BST molecule. Due to differences in the molecular mass it is possible to distinguish between rBST and natural BST. Milk production in various countries, its processing into dairy products and its consumption are summarized in Tables 10.1–10.3.

10.1.1 Physical and Physico-Chemical Properties

Milk is a white or yellow-white, opaque liquid. The color is influenced by scattering and absorp-

tion of light by milk fat globules and protein micelles. Therefore, skim milk also retains its white color. A yellowish, i. e. yellow-green, color is derived from carotene (ingested primarily during pasture grazing) present in the fat phase and from riboflavin present in the aqueous phase. Milk tastes mildly sweet, while its odor and flavor are normally quite faint.

Milk fat occurs in the form of droplets or globules, surrounded by a membrane and emulsified in milk serum (also called whey). The fat globules (called cream) separate after prolonged storage or after centrifugation. The fat globules float on the skim milk. Homogenization of milk so finely divides and emulsifies the fat globules that cream separation does not occur even after prolonged standing.

Proteins of various sizes are dispersed in milk serum. They are called micelles and consist mostly of calcium salts of casein molecules. Furthermore, milk contains lipoprotein particles, also called milk microsomes, which consist of the residues of cell membranes, microvilli, etc., as well as somatic cells, which are mainly leucocytes (10^8 /l of milk). Some of the properties of the main structural elements of milk are listed in Table 10.4.

Various proteins, carbohydrates, minerals and other ingredients are solubilized in milk serum. The specific density of milk decreases with increasing fat content, and increases with increasing amounts of protein, milk sugar and salts. The specific density of cow’s milk ranges from 1.029 to 1.039 (15 °C). Defatted (skim) milk has a higher specific density than whole milk. From the relationships given by *Fleischmann*:

$$m = 1.2f + \frac{266.5(s - 1)}{s} \quad (10.1)$$

and by *Richmond*:

$$m = 0.25s + 1.21f + 0.66 \quad (10.2)$$

Table 10.1. Production of milk, 2006 (1000 t)

Continent	Cow milk	Buffalo milk	Sheep milk	Goat milk
World	549,693	80,094	8723	13,801
Africa	24,674	2300	1719	3129
America, Central-	14,179	–	–	–
America, North-	90,564	–	–	–
America, South- and Caribbean	66,030	–	36	164
Asia	134,170	77,571	4006	7821
Europe	209,441	222	2963	2479
Oceania	24,814	–	–	–

Country	Cow milk	Country	Buffalo milk	Country	Sheep milk
USA	82,463	India	52,100	China	1091
India	39,775	Pakistan	21,136	Turkey	790
China	32,249	China	2850	Greece	752
Russian Fed.	31,074	Egypt	2300	Syria	604
Germany	28,453	Nepal	927	Italy	554
Brazil	25,333	Iran	232	Romania	545
France	24,195	Italy	215	Iran	534
UK	14,577	Myanmar	171	Sudan	487
New Zealand	14,498	Turkey	38	Spain	403
Ukraine	12,988	Viet Nam	31	France	263
Poland	11,982	Σ (%) ^a	100	Algeria	210
Italy	11,013			Mali	128
Netherlands	10,532			Bulgaria	108
Australia	10,250			Portugal	100
Mexico	10,029			Σ (%) ^a	75
Turkey	10,026				
Pakistan	9404				
Japan	8134				
Argentina	8100				
Canada	8100				
Colombia	6770				
Σ (%) ^a	75				

Country	Goat milk
India	3790
Sudan	1519
Bangladesh	1416
Pakistan	676
France	583
Greece	511
Spain	423
Iran	365
China	262
Ukraine	258
Russian Fed.	256
Turkey	254
Σ (%) ^a	75

^a World production = 100%.

Table 10.2. Production of dairy products in 2004 (1000 t)

Continent	Cheese	Butter ^a	Condensed milk	Whole milk powder	Skim milk powder ^b	Whey powder
World	17,824	7968	3892	2702	3455	2038
Africa	915	226	64	21	11	2
America, North-, Central-	4944	646	1112	140	796	542
America, South-	668	191	377	768	64	–
Asia	1090	3678	559	83	239	4
Europe	9558	2622	1760	946	1699	1386
Oceania	649	605	21	744	647	105

Country	Cheese	Country	Butter ^a	Country	Condensed milk
USA	4357	India	2500	USA	797
Germany	1852	Pakistan	557	Germany	505
France	1840	USA	525	The Netherlands	291
Italy	1320	New Zealand	473	Peru	274
The Netherlands	670	Germany	440	Russian Fed.	193
Egypt	661	France	420	Thailand	179
Poland	520	Russian Fed.	262	Malaysia	164
Russian Fed.	483	Poland	180	Mexico	158
UK	370	UK	160	UK	139
Australia	364	Iran	150	China	114
Argentina	360	Ireland	142	Ukraine	80
Canada	360	Australia	130	Canada	78
Denmark	335	Italy	125	Σ (%) ^c	76
Σ (%) ^c	76	Σ (%) ^c	76		

Country	Whole milk powder	Country	Skim milk powder ^b	Country	Whey powder
New Zealand	557	USA	674	France	610
Brazil	420	New Zealand	425	USA	493
France	220	France	271	Germany	262
Australia	187	Germany	250	The Netherlands	219
Argentina	165	Russian Fed.	243	Australia	82
The Netherlands	112	Australia	222	UK	56
Mexico	105	Japan	180	Canada	49
UK	90	Poland	140	Denmark	39
Russian Fed.	85	Ukraine	117	Finnland	32
Denmark	80	Canada	102	Ireland	30
Σ (%) ^c	75	Σ (%) ^c	76	Σ (%) ^c	92

^a Including fat from buffalo milk (ghee)^b Including butter-milk powder^c World production = 100%

the dry matter content of milk, *m*, in percent, can be calculated from the percent fat content (*f*), knowing the specific density (*s*).

The freezing point of milk is -0.53 to -0.55 °C. This rather constant value is a suitable test for detection of watering of milk.

The pH of fresh milk is 6.5–6.75, while the acid degree according to *Soxhlet–Henkel* (°SH) is 6.5–7.5.

The refractive index (n_D^{20}) is 1.3410–1.3480, and the specific conductivity at 25 °C is 4 – 5.5×10^{-3} ohm⁻¹cm⁻¹.

Table 10.3. Consumption of milk and dairy products in FR Germany (in kg/capita and year)

	1996	2003	2005
Consumer milk	66.7	66	67
Fresh milk products (without yoghurt)	9.9	12.2	12
Yoghurt	13.1	15.3	16.8
Cream and cream products	7.6	7.4	7.4
Butter	7.3	6.6	6.5

The measurement of redox potentials of milk and its products can also be of value. The redox potential is +0.30 V for raw and +0.10 V for pasteurized milk, +0.05 V for processed cheese, −0.15 V for yoghurt and −0.30 V for Emmental cheese.

10.1.2 Composition

The composition of dairy cattle milk varies to a fairly significant extent. Table 10.5 provides some data. In all cases water is the main ingredient of milk at 63–87%. In the following sections, only cow's milk will be dealt with in detail since it is the main source of our dairy foods.

10.1.2.1 Proteins

In 1877 *O. Hammarsten* distinguished three proteins in milk: casein, lactalbumin and lactoglobulin. He also outlined a procedure for their separation: skim milk is diluted then acidified with acetic acid. Casein flocculates, while the

Table 10.5. Composition of human milk and milk of various mammals (%)

Milk	Protein	Casein	Whey protein	Sugar	Fat	Ash
Human	0.9 ^a	0.4	0.5	7.1	4.5	0.2
Cow (bovine)	3.2	2.6	0.6	4.6	3.9	0.7
Donkey	2.0	1.0	1.0	7.4	1.4	0.5
Horse	2.5	1.3	1.2	6.2	1.9	0.5
Camel	3.6	2.7	0.9	5.0	4.0	0.8
Zebu	3.2	2.6	0.6	4.7	4.7	0.7
Yak	5.8			4.6	6.5	0.9
Buffalo	3.8	3.2	0.6	4.8	7.4	0.8
Goat	3.2	2.6	0.6	4.3	4.5	0.8
Sheep	4.6	3.9	0.7	4.8	7.2	0.9
Reindeer	10.1	8.6	1.5	2.8	18.0	1.5
Cat	7.0	3.8	3.2	4.8	4.8	0.6
Dog	7.4	4.8	2.6			
Rabbit	10.4					

^a After the 15-th day of the breast feeding period the protein content is increased to 1.6%.

whey proteins stay in solution. This established a specific property of casein: it is insoluble in weakly acidic media. It was later revealed that the milk protein system is much more complex. In 1936 *Pedersen* used ultra-centrifugation to demonstrate the nonhomogeneity of casein, while in 1939 *Mellander* used electrophoresis to prove that casein consists of three fractions, i.e. α -, β - and γ -casein. The most important proteins of milk are listed in Table 10.7. The casein fraction forms the main portion. Major constituents of whey proteins, β -lactoglobulin A and B and α -lactalbumin, can be differentiated genetically. Other protein constituents, e.g., enzymes, are present in much lower quantities; they are not listed in Table 10.7.

Table 10.4. Main structural elements of milk

Name	Type of dispersion	Percentage	Number (1 ⁻¹)	Diameter (mm)	Surface (m ² /1 milk)	Specific density ^a (g/ml)
Fat globules	Emulsion	3.8	10 ¹³	100–10,000	70	0.92
Casein micelles	Suspension	2.8	10 ¹⁷	10–300	4000	1.11
Globular proteins (whey proteins)	Colloidal solution	0.6	10 ²⁰	3–6	5000	1.34
Lipoprotein particles	Colloidal suspension	0.01	10 ¹⁷	10	10	1.10

^a 20 °C.

Table 10.6. Amino acid composition (g AA/100 g protein) of the total protein, casein, and whey protein of bovine milk

Amino acid	Total protein	Casein	Whey protein
Alanine	3.7	3.1	5.5
Arginine	3.6	4.1	3.3
Aspartic acid	8.2	7.0	11.0
Cystine	0.8	0.3	3.0
Glutamic acid	22.8	23.4	15.5
Glycine	2.2	2.1	3.5
Histidine	2.8	3.0	2.4
Isoleucine	6.2	5.7	7.0
Leucine	10.4	10.5	11.8
Lysine	8.3	8.2	9.6
Methionine	2.9	3.0	2.4
Phenylalanine	5.3	5.1	4.2
Proline	10.2	12.0	4.4
Serine	5.8	5.5	5.5
Threonine	4.8	4.4	8.5
Tryptophan	1.5	1.5	2.1
Tyrosine	5.4	6.1	4.2
Valine	6.8	7.0	7.5

The amino acid composition of the total protein, casein, and whey protein of bovine milk is presented in Table 10.6.

10.1.2.1.1 Casein Fractions

The main constituents of this milk protein fraction have been fairly well investigated.

Their amino acid sequences are summarized in Table 10.8. Data showing the genetic variations are provided in Table 10.9. Caseins are not denaturable because of the lacking tertiary structure.

α_s -Caseins. The B variant of α_{s1} -casein consists of a peptide chain with 199 amino acid residues and has a molecular weight of 23 kdal. The sequence contains 8 phosphoserine residues, 7 of which are localized in positions 43–80, and these positions have an additional 12 carboxyl groups. Thus these positions are extremely polar acidic segments along the peptide chain. Proline is uni-

Table 10.7. Bovine milk proteins

Fraction	Genetic variants	Portion ^a	Isoionic point	Molecular weight ^b (kdal)	Phosphorus content (%)
<i>Caseins</i>		80	—	—	0.9
α_{s1} -Casein	A, B, C, D, E	34	4.92–5.35	23.6 ^f	1.1
α_{s2} -Casein	A, B, C, D	8		25.2 ^g	1.4
κ -Casein	A, B, C, E	9	5.77–6.07	19 ^h	0.2
β -Casein	A ¹ , A ² , A ³ , B, C, D, E	25	5.20–5.85	24	0.6
γ -Casein		4	5.8–6.0	12–21	0.1
γ_1 -Casein	A ¹ , A ² , A ³ , B			20.5	
γ_2 -Casein	A ¹ /A ² , A ³ , B			11.8	
γ_3 -Casein	A ¹ /A ² /A ³ , B			11.6	
<i>Whey proteins</i>		20	—	—	
β -Lactoglobulin	A, B, C, D, E, F, G	9	5.35–5.41	18.3	
α -Lactalbumin	A, B, C	4	4.2–4.5 ^e	14.2	
Serum albumin	A	1	5.13	66.3	
Immunoglobulin		2			
IgG1			5.5–6.8	162	
IgG2			7.5–8.3	152	
IgA			—	400 ^c	
IgM			—	950 ^d	
FSC(s) ⁱ				80	
Proteose-Peptide		4	3.3–3.7	4–41	

^a As % of skim milk total protein, ^b monomers, ^c dimer, ^d pentamer, ^e isoelectric point, ^f Variant B, ^g Variant A,

^h Variant A², ⁱ Free secretory component

Table 10.8. Amino acid sequences of bovine milk proteins

<i>α_{s1}-Casein B-8P</i>																			
R	P	K	H	P	I	K	H	Q	G	L	P	Q	E	V	L	N	E	N	L
L	R	F	F	V	A	P	F	P	Q	V	F	G	K	E	K	V	N	Q	L
S	K	D	I	G	S ^a	E	S ^a	T	E	D	Q	A	M	E	D	I	K	E	M
E	A	E	S ^a	I	S ^a	S ^a	S ^a	E	E	I	V	P	N	S ^a	V	Q	E	K	H
I	Q	K	E	D	V	P	S	E	R	Y	L	G	Y	L	E	Q	L	L	R
L	K	K	Y	K	V	P	Q	L	E	I	V	P	N	S ^a	A	E	E	R	L
H	S	M	K	E	G	I	H	A	Q	Q	K	E	P	M	I	G	V	N	Q
E	L	A	Y	F	Y	P	E	L	F	R	Q	F	Y	Q	L	D	A	Y	P
S	G	A	W	Y	Y	V	P	L	G	T	Q	Y	T	D	A	P	S	F	S
D	I	P	N	P	I	G	S	E	N	S	E	K	T	T	M	P	L	W	
<i>α_{s2}-Casein A-11P</i>																			
K	N	T	M	E	H	V	S ^a	S ^a	S ^a	E	E	S	I	I	S ^a	Q	E	T	Y
K	Q	E	K	N	M	A	I	N	P	S	K	E	N	L	C	S	T	F	C
K	E	V	V	R	N	A	N	E	E	E	Y	S	I	G	S ^a	S ^a	E	E	
S ^a	A	E	V	A	T	E	E	V	K	I	T	V	D	D	K	H	Y	Q	K
A	L	N	E	I	N	E	F	Y	Q	K	F	P	Q	Y	L	Q	Y	L	Y
Q	G	P	I	V	L	N	P	W	D	Q	V	K	R	N	A	V	P	I	T
P	T	L	N	R	E	Q	L	S ^a	T	S ^a	E	E	N	S	K	K	T	V	D
M	E	S ^a	T	E	V	F	T	K	K	T	K	L	T	E	E	E	K	N	R
L	N	F	L	K	K	I	S	Q	R	Y	Q	K	F	A	L	P	Q	Y	L
K	T	V	Y	Q	H	Q	K	A	M	K	P	W	I	Q	P	K	T	K	V
I	P	Y	V	R	Y	L													
<i>β-Casein A²-5P</i>																			
R	E	L	E	E	L	N	V	P	G	E	I	V	E	S ^a	L	S ^a	S ^a	S ^a	E
E	S	I	T	R	I	N	K	K	I	E	K	F	Q	S ^a	E	E	Q	Q	Y
T	E	D	E	L	Q	D	K	I	H	P	F	A	Q	T	Q	S	L	V	Y
P	F	P	G	P	I	P	N	S	L	P	Q	N	I	P	P	L	T	Q	T
P	V	V	V	P	P	F	L	Q	P	E	V	M	G	V	S	K	V	K	E
A	M	A	P	K	H	K	E	M	P	F	P	K	Y	P	V	Q	P	F	T
E	S	Q	S	L	T	L	T	D	V	E	N	L	H	L	P	P	L	L	L
Q	S	W	M	H	Q	P	H	Q	P	L	P	P	T	V	M	F	P	P	Q
S	V	L	S	L	S	Q	S	K	V	L	P	V	P	E	K	A	V	P	Y
P	Q	R	D	M	P	I	Q	A	F	L	L	Y	Q	Q	P	V	L	G	P
V	R	G	P	F	P	I	I	V											
<i>κ-Casein B-1P</i>																			
Z ^d	E	Q	N	Q	E	Q	P	I	R	C	E	K	D	E	R	F	F	S	D
K	I	A	K	Y	I	P	I	Q	Y	V	L	S	R	Y	P	S	Y	G	L
N	Y	Y	Q	Q	K	P	V	A	L	I	N	N	Q	F	L	P	Y	P	Y
Y	A	K	P	A	A	V	R	S	P	A	Q	I	L	Q	W	Q	V	L	S
D	T	V	P	A	K	S	C	Q	A	Q	P	T	T	M	A	R	H	P	H
P	H	L	S	F	M	A	I	P	P	K	K	N	Q	D	K	T	E	I	P
T	I	N	T	I	A	S	G	E	P	T ^b	S	T ^b	P	T ^b	I	E	A	V	E
S	T	V	A	T	L	E	A	S ^a	P	E	V	I	E	S	P	P	E	I	N
T	V	Q	V	T	S	T	A	V											

formly distributed along the chain and apparently to a great extent hinders the formation of a regular structure. A portion of the chain, up to 30%,

is assumed to have regular conformations. Amino acid residues 100–199 are distinctly apolar and are responsible for strong association tendencies,

Table 10.8. continued

<i>α-Lactalbumin B^c</i>																			
E	Q	L	T	K	C	E	V	F	R	E	L	K	D	L	K	G	Y	G	G
V	S	L	P	E	W	V	C	T	T	F	H	T	S	G	Y	D	T	E	A
I	V	E	N	N	Q	S	T	D	Y	G	L	F	Q	I	N	N	K	I	W
C	K	N	D	Q	D	P	H	S	S	N	I	C	N	I	S	C	D	K	F
L	N	N	D	L	T	N	N	I	M	C	V	K	K	I	L	D	K	V	G
I	N	Y	W	L	A	H	K	A	L	C	S	E	K	L	D	Q	W	L	C
E	K	L																	
<i>β-Lactoglobulin B^c</i>																			
L	I	V	T	Q	T	M	K	G	L	D	I	Q	K	V	A	G	T	W	Y
S	L	A	M	A	A	S	D	I	S	L	L	D	A	Q	S	A	P	L	R
V	Y	V	E	E	L	K	P	T	P	E	G	D	L	E	I	L	L	Q	K
W	E	N	G	E	C	A	Q	K	K	I	I	A	E	K	T	K	I	P	A
V	F	K	I	D	A	L	N	E	N	K	V	L	V	L	D	T	D	Y	K
K	Y	L	L	F	C	M	E	N	S	A	E	P	E	Q	S	L	A	C	Q
C	L	V	R	T	P	E	V	D	D	E	A	L	E	K	F	D	K	A	L
K	A	L	P	M	H	I	R	L	S	F	N	P	T	Q	L	E	E	Q	C
H	I																		

^a The serine residue is phosphorylated.

^b These threonine residues can be glycosylated.

^c Disulfide bonds: 6–120, 28–111, 61–77, 73–91.

^d Pyrrolidone carboxylic acid.

^e Disulfide bonds: 66–160 and apparently either 106–119 or 106–121. Accordingly, the free thiol group is either Cys-119 or Cys-121.

which are limited by the repulsing forces of phosphate groups. In the presence of Ca^{2+} ions, in the levels found in milk, α_{s1} -casein forms an insoluble Ca-salt. In the A variant of the molecule, amino acid residues 14–26 are missing; in the C variant the glutamic acid in position 192 (Glu-192) is replaced by Gly-192; and in the D variant Pth-53 (phosphothreonine) replaces Ala-53.

α_{s2} -Casein (M_r 25,000) consists of 207 amino acid residues, has a pronounced dipolar structure with a concentration of anionic groups in the region of the N-terminus and cationic groups in the region of the C-terminus. It contains 11 phosphoserine and 2 cysteine residues and is even more easily precipitable with Ca^{2+} than α_{s1} -casein. Other proteins, previously known as α_{s3} -, α_{s4} -, α_{s5} -, and α_{s6} -caseins, appear to be members of the α_{s2} family and to differ in the degree of phosphorylation. Dimers linked via disulfide bridges also appear to be present.

β-Caseins. The A² variant is a peptide chain consisting of 209 residues and has a molecular weight of 24.0 kdal. Five phosphoserine residues

are localized in positions 1–40; these positions contain practically all of the ionizing sites of the molecule. Positions 136–209 contain mainly residues with apolar side chains. On the whole, β-casein is the most hydrophobic casein. The molecule has a structure with a “polar head” and an “apolar tail”, thus resembling a “soaplike” molecule. Indeed, CD measurements have shown that β-casein contains about 9% of α-helix structure and about 25% of β-structure. An increase in temperature results in an increase in the β-structure at the cost of the aperiodic part. The self-association of β-casein is an endothermic process. Like α_{s1} -casein, β-casein contains no cysteine. The protein precipitates in the presence of Ca^{2+} ions at the levels found in milk. However, at temperatures at or below 1 °C the calcium salt is quite soluble.

κ-Caseins. The B variant consists of a peptide chain with 169 residues and has a molecular weight of 18 kdal. The monomer, which contains 1 phosphoserine and 2 cysteine residues, is accessible only under reducing conditions.

Table 10.9. Amino acid sequences^a of genetic variants of bovine milk proteins

Protein	Variant	Frequency ^b	Positions of the substitutions									
α_{s1} -Casein (199 AS)	A	s	14–26 are lacking		53		59		192			
	B	w			Ala		Glu		Glu			
	C	i							Gly			
α_{s2} -Casein (207 AS)	D	s			ThrP							
	E	s					Lys		Gly			
	A		33	47	50–58	130						
	B		Glu	Ala		Thr						
	C		Gly	Thr		Ile						
β -Casein (209 AS)	D				lacking							
	A ¹		18	35	36	37	67	106	122			
	A ²	w, i	SerP	SerP	Glu	Glu	Pro	His	Ser			
	A ³							Gln				
	B	s					His		Arg			
	C	s		Ser		Lys	His					
	D	s	Lys									
κ -Casein (169 AS)	E	s				Lys						
	A	x		97	136	148	155					
	B	w, i		Arg	Thr	Asp	Ser					
	C				Ile	Ala						
α -Lactalbumin (123 AS)	E						Gly					
	A	i	10									
β -Lactoglobulin (162 AS)	B	w	Gln									
	A	x	45	50	59	64	78	118	130	158		
	B	w, i	Glu	Pro	Gln	Asp	Ile	Val	Asp	Gln		
	C				His	Gly		Ala				
	D		Gln									
	E										Gly	
	F			Ser					Tyr	Gly	Gly	
β -Lactoglobulin (162 AS)	G						Met			Gly		

^a cf. Table 10.8.^b w: predominant in the western world (*Bos taurus*), i: predominant in India (*Bos indicus*, *Bos grunniens*), s: rare, x: not predominant, but not rare.

Normally, κ -casein occurs as a trimer or as a higher oligomer in which the formation of disulfide bonds is probably involved. The protein contains varying amounts of carbohydrates (average values: 1% galactose, 1.2% galactosamine, 2.4% N-acetyl neuramic acid) that are bound to the peptide chain through Thr-131, 133, 135 or (in variant A) 136. κ -Casein is separated electrophoretically into various components that

have the same composition of amino acids, but differ in their carbohydrate moiety, e. g., per protein molecule they contain 0–3 moles N-acetyl neuramic acid, 0–4 moles galactose and 0–3 moles galactosamine. Three different glycosyl residues could be isolated, one of which has the structure shown in Formula 10.3.

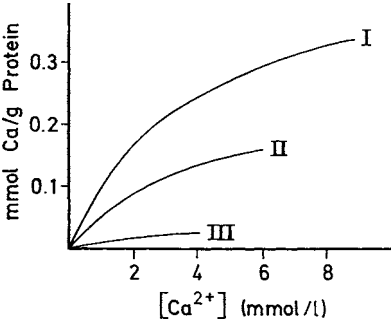


Fig. 10.1. Calcium binding by I: α_{s1} -casein (0.38), II: β -casein (0.21) and III: κ -casein (0.05). The bound phosphate residues in mmol/g of casein are given in brackets (according to Walstra and Jenness, 1984)

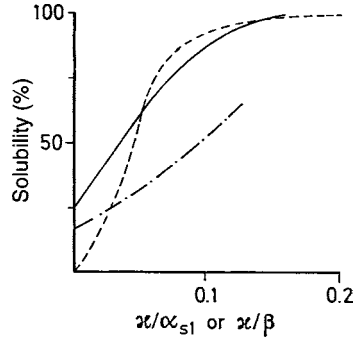
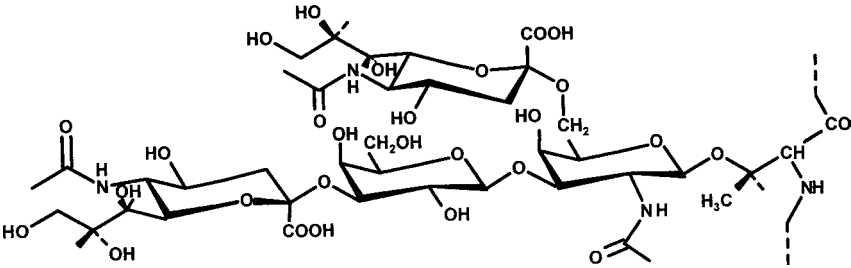


Fig. 10.2. Influence of κ -casein on the solubility of α_{s1} -casein. (−2.5 mg/ml) and β -casein (−1.5 mg/ml; −6 mg/ml) at pH 7.0, 30 °C, 100 mmol/l CaCl_2 (according to Walstra and Jenness, 1984)

In the other two oligosaccharide units, one of the two N-acetylneuraminic acid residues is lacking in each case. κ -Casein is the only main constituent of casein which remains soluble in the presence of Ca^{2+} ions in the concentrations found in milk (Fig. 10.1). Aggregation of α_{s1} - and β -caseins with κ -casein prevents their coagulation in the presence of Ca^{2+} ions (Fig. 10.2). This property of κ -casein is of utmost importance for formation and maintenance of stable casein complexes and casein micelles, as occur in milk. Chymosin (rennet, rennin cf. 1.4.5.2) selectively cleaves the peptide chain of κ -casein at $-\text{Phe}^{105}-\text{Met}^{106}-$ into two fragments: para- κ -casein and a glycopeptide (Pyg = pyroglutamic acid, i.e. pyrrolidone carboxylic acid):

The released glycopeptide is soluble, while para- κ -casein precipitates in the presence of Ca^{2+} ions. In this way κ -casein loses its protective effect; the casein complexes and casein micelles coagulate (curdle formation) from the milk. The specificity of rennin is high, as is shown in Table 10.10. If Met^{106} in κ -casein is replaced with Phe^{106} by genetic engineering techniques, the rate of catalysis is increased by 80%. The sugar moiety of κ -casein is not essential for rennin action, nor for the stabilizing property of its protein portion. However, the sugar moiety delays protein cleavage by rennin. Also, it appears that the stability of α_s - and κ -casein mixtures in the presence of Ca^{2+} ions is influenced by the carbohydrate content of κ -casein.



(10.3)

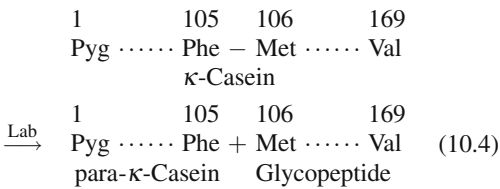


Table 10.10. Chymosin specificity: relative rate of hydrolysis of peptides from the κ -casein amino acid sequence

Substrate		V_{rel}^a
105	106	
Phe-Met		0.00
104	108	
Ser-Phe-Met-Ala-Ile		0.04
	109	
Ser-Phe-Met-Ala-Ile-Pro		0.11
103		
Leu-Ser-Phe-Met-Ala-Ile		21.6
102		
His-Leu-Ser-Phe-Met-Ala-Ile		31
	110	
Leu-Ser-Phe-Met-Ala-Ile-Pro-Pro		100
101		
Pro-His-Leu-Ser-Phe-Met-Ala-Ile		100
98	112	
His-Pro-His-Pro-His-Leu-Ser-Phe-Met-Ala-Ile-Pro-Pro-Lys-Lys		2500

^a Relative rate: k_{cat}/K_m .

In the C variant of κ -casein, Arg⁹⁷ is replaced with His⁹⁷ (Table 10.9), which has a weaker positive charge. As a result, chymosin is not as strongly bound as in the case of the B variant; the rate of catalysis decreases. Therefore, C variant milk is less suitable for the production of sweet-milk cheese than B variant milk.

γ -Caseins. These proteins are degradation products of the β -caseins, formed by milk proteases, e. g., γ_1 -casein is obtained by cleavage of the residues 1–28. The peptide released is identical to the proteose-peptone PP8F which has been found in milk. Correspondingly, γ_2 - and γ_3 -caseins are formed by hydrolysis of the amino acid residues 1–105 and 1–107 respectively. According to more recent nomenclature recommendations, β -casein fragments should be described by the position numbers. Thus, γ_1 -casein from any β -casein variant X is called, e. g., β -casein X (f29–209) and the corresponding proteose peptone PF8F β -casein X (f1–28).

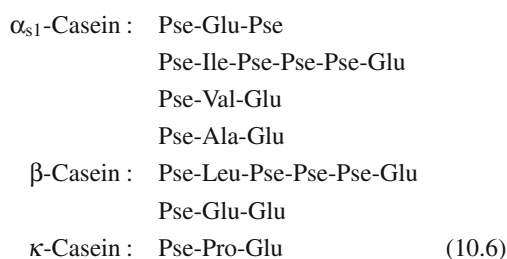
λ -Caseins. The λ -casein fraction consists mainly of fragments of the α_{s1} -caseins. In vitro the α -caseins are formed by incubation of the α_{s1} -caseins with bovine plasmin.

The molar ratio of the main components $\alpha_{s1}/\beta + \gamma/\kappa/\alpha_{s2}$ is on an average 8/8/3/2. All casein forms contain phosphoric acid, which always occurs in a tripeptide sequence pattern (Pse = phos-

phoserine):



in which X is any amino acid, including phosphoserine and glutamic acid. Examples are:



Most probably this regular pattern originates from the action of a specific protein kinase. The various distribution of polar and apolar groups of the individual proteins outlined above are summarized in Table 10.11. The hydrophobicity values listed are average hydrophobicity values $\bar{H}$ of the amino acid side chains present in the sequence of the given segments, and are calculated as follows:

A measure of the hydrophobicity of a compound is the free energy, F_t , needed to transfer the compound from water into an organic solvent, and is given as the ratio of the compound's solubility in water (N_w , as mole fraction) and in the organic solvent (N_{org} , as mole fraction), involving the ac-

Table 10.11. Distribution of amino acid residues with ionizing side chains (net charge) and with nonpolar side chains (hydrophobicity) in α_{s1} -casein and β -casein

Residue	α_{s1} -Casein		Residue	β -Casein	
	1	2		1	2
1–40	+3	1340	1–43	–16	783
41–80	–22.5	641	44–92	–3.5	1429
81–120	0	1310	93–135	+2	1173
121–160	–1	1264	136–177	+3	1467
161–199	–2.5	1164	178–209	+2	1738

1 Net charge.
2 Hydrophobicity $\bar{H}$ (Cal/mole; cf. text).

tivity coefficients (γ_w, γ_{org}):

$$\Delta F_t = RT \ln \frac{N_w \cdot \gamma_w}{N_{org} \cdot \gamma_{org}} \quad (10.7)$$

The corresponding free energy of transfer of the side chain of an amino acid $H\Phi_i$ is obtained from the following relationship:

$$H\Phi_i = \Delta F_t(\text{amino acid } i) - \Delta F_t(\text{glycine})$$

The average hydrophobicity of a sequence segment of a polypeptide chain with n amino acid residues is then:

$$\bar{H} = \frac{\sum H\Phi_i}{n} \quad (10.8)$$

The higher the $H\Phi_i$, i. e. $\bar{H}$, the higher is the hydrophobicity of individual side chains, i. e. the sequence segment. Data provided in Table 10.11 are related to the ethanol/water system.

10.1.2.1.2 Micelle Formation

Only up to 10% of the total casein fraction is present as monomers. They are usually designated as serum caseins and the concentration ratio $c_\beta > c_\kappa > c_{\alpha_{s1}}$ is quite valid. However, the main portion is aggregated to casein complexes and casein micelles. This aggregation is regulated by a set of parameters, as presented in Fig. 10.3. Dialysis of casein complexes against a chelating agent might shift the equilibrium completely to

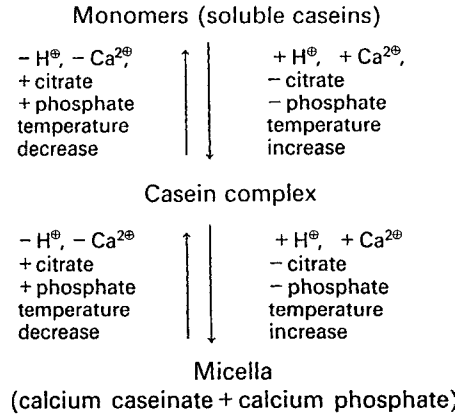


Fig. 10.3. Casein complex and casein micelle formation

monomers, while against high Ca^{2+} ion concentrations the shift would be to large micelles.

From Fig. 10.4 it follows that the diameter of the micelles in skim milk varies from 50–300 nm, with a particle distribution peak at 150 nm. Using an average diameter of 140 nm, the micelle volume is $1.4 \times 10^6 \text{ nm}^3$ and the particle weight is $10^7\text{--}10^9 \text{ dal}$. This corresponds to 25,000 monomers per micelle. Casein micelles are substantially smaller than fat globules, which have diameters between 0.1–10 μm . Scanning electron micrographs of micelles are shown in Fig. 10.5 and compositional data are provided in Table 10.12.

The ratio of monomers in micelles varies to a great extent (Table 10.13), depending on dairy cattle breed, season and fodder, and is influenced also by micellar size (Table 10.14). The micelles are not tightly packed and so are of variable density. They are strongly solvated

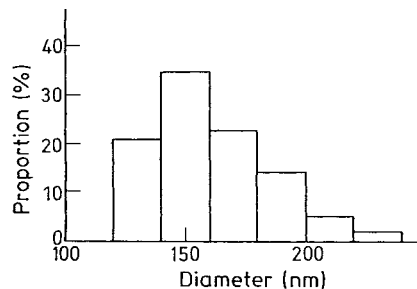


Fig. 10.4. Particle size distribution of casein micelles in skim milk (fixation with glutaraldehyde)

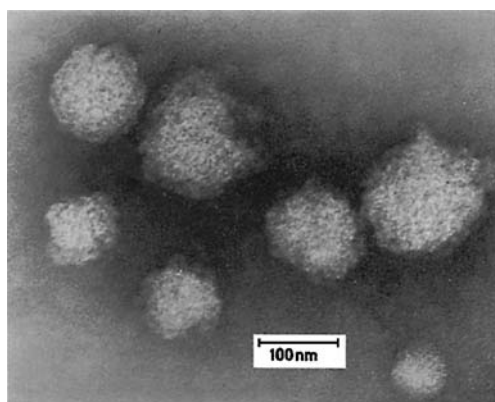


Fig. 10.5. Electron micrograph of the casein micelles in skim milk (according to *Webb*, 1974). The micelles are fixed with glutaraldehyde and then stained with phosphomolybdic acid

Table 10.12. Composition of casein micelles (%)

Casein	93.2	Phosphate	
Ca	2.9	(organic)	2.3
Mg	0.1	Phosphate	
Na	0.1	(inorganic)	2.9
K	0.3	Citrate	0.4

Table 10.13. Typical distribution of components in casein micelles

Component	Ratio numbers			
α_{s1}	3	6	9	12
β	1	1	4	4
γ		1	1	1
κ	1	3	3	3

(1.9 g water/g protein) and hence are porous. The monomers are kept together with:

- Hydrophobic interactions that are minimal at a temperature less than 5 °C.
- Electrostatic interactions, mostly as calcium or calcium phosphate bridges between phosphoserine and glutamic acid residues (Fig. 10.6).
- Hydrogen bonds.

On a molecular level different micelle models have been proposed which to a certain extent explain the experimental findings. The most probable model is shown in Fig. 10.7. This model comprises subunits (submicelles, $M_r \sim 760,000$)

Table 10.14. Composition and size of casein micelles isolated by centrifugation

Centrifugation time (min) ^a	Composition of the sediment(%)			
	α_{s1}	β	κ	Others
0 ^b	50	32	15	3
0–7.5	47	34	16	3
7.5–15	46	32	18	4
15–30	45	31	20	4
39–60	42	29	26	3
Serum casein	39	23	33	5

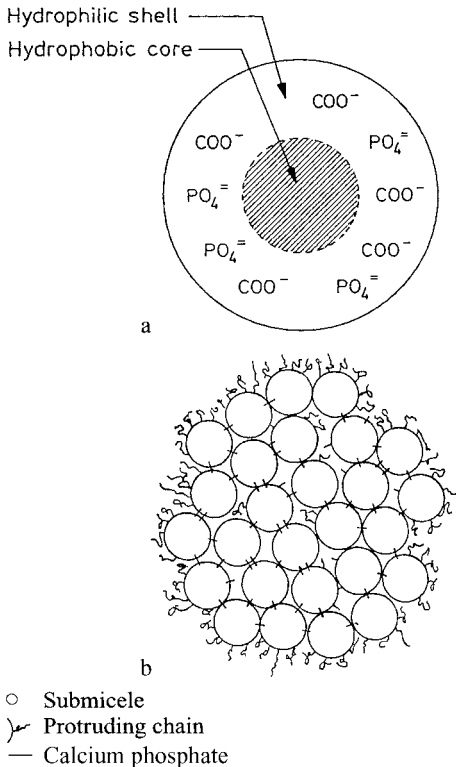
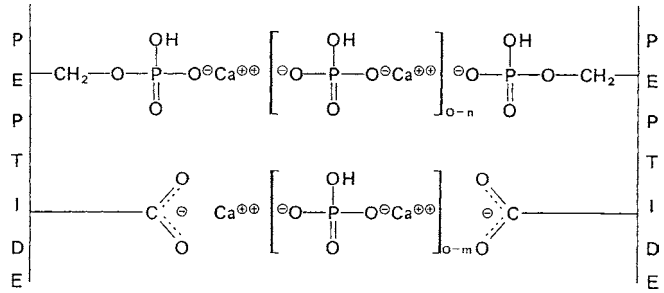
^a Centrifugation speed $10^5 \times g$.

^b Isoelectric casein.

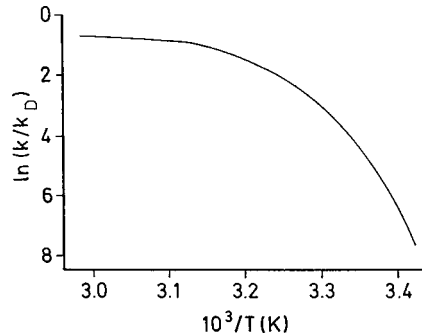
which consist of ca. 30 different casein monomers and aggregate to large micelles via calcium phosphate bridges. Two types of subunits apparently exist: one type contains κ -casein and the other does not. The κ -casein molecules are arranged on the surface of the corresponding submicelles. At various positions, their hydrophilic C-termini protrude like hairs from the surface, preventing aggregation. Indeed, aggregation of the submicelles proceeds until the entire surface of the forming micelle is covered with κ -casein, i.e., covered with “hair”, and, therefore, exhibits steric repulsion. The effective density of the hair layer is at least 5 nm. A small part of the κ -casein is also found inside the micelle.

10.1.2.1.3 Gel Formation

The micelle system, can be destabilized by the action of rennin or souring. Rennin attacks κ -casein, eliminating not only the C-terminus in the form of the soluble glycopeptide 106–169, but also the cause of repulsion. The remaining paracasein micelles first form small aggregates with an irregular and often long form, which then assemble with gel formation to give a three dimensional network with a pore diameter of a few μm . The fat globules present are included in this network with pore enlargement. It is assumed that dynamic equilibria exist between casein monomers and submicelles, dissolved and bound calcium phosphate, and submicelles and micelles.

Fig. 10.6. Peptide chain bridging with calcium ions**Fig. 10.7.** Schematic model of a casein micelle; (a) a subunit consisting of α_{s1} -, β -, γ -, κ -caseins, (b) Micelle made of subunits bound by calcium phosphate bridges (according to Webb, 1974)

The rate of gel formation increases with increasing temperature (Fig. 10.8). It is slow at $T < 25^\circ\text{C}$ and proceeds almost under diffusion control at $T \sim 60^\circ\text{C}$. It follows that hydrophobic interactions, especially due to the very hydrophobic para- κ -casein remaining on the surface after the action of rennin, are the driving force for

**Fig. 10.8.** Temperature dependency of the aggregation rate of para-casein micelles (rate constant k in fractions of the diffusion-controlled rate k_D ; according to Dalgleish, 1983)

gel formation. In addition, other temperature-dependent reactions play a role, like the binding of calcium ions and of β -casein to the micelles, and the change in solubility of colloidal calcium phosphate.

Acid coagulation of casein is also definitely caused by hydrophobic interactions, as shown by the dependency of the coagulation rate on the temperature and pH value (Fig. 10.9). On acidification, the micelle structure changes due to the migration of calcium phosphate and monomeric casein. Since the size of the micelle remains practically constant, this migration of components must be associated with swelling. During coagulation, dissolved casein reassociates with the micelles, forming a gel network.

The gel structure can be controlled via changes in the hydrophobicity of the micelle surface. A decrease in hydrophobicity is possible, e. g., by heating milk ($90^\circ\text{C}/10\text{ min}$). Covalent bonding of denatured β -lactoglobulin to κ -casein (cf. 10.1.3.5) occurs, burying hydrophobic groups.

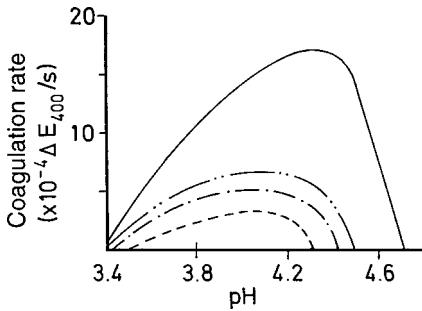


Fig. 10.9. Rate of coagulation of casein micelles as a function of temperature and pH value (— 25 °C, ···· 15 °C, - · - · 10 °C, --- 5 °C, according to *Bringe, Kinsella, 1986*)

Due to weaker interactions, stable, rigid gels with a chain-like structure are formed on acidification. These gels exhibit no syneresis and are desirable, e.g., in yoghurt (10.2.1.2). Figure 10.10 shows that the firmness of stiff yoghurt is highest when the denaturation of β -lactoglobulin is 90–99%. If this rate of denaturation is achieved at lower temperatures (e.g., 85 °C), gels are formed that are more rigid and coarser than those formed by heating to higher temperatures (e.g., 130 °C), which results in a soft, smooth gelatinous mass. The gel stability of whole-milk yoghurt is lower than that of skim-milk yoghurt because the protein network is interrupted by included fat globules.

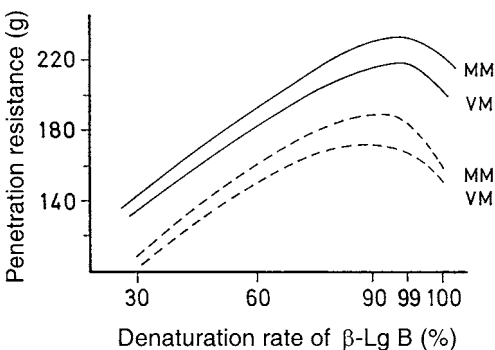


Fig. 10.10. Firmness of yoghurt as a function of the rate of denaturation of β -lactoglobulin B (the final value of the penetration resistance of a conical test piece in stiff yoghurt is given; heating temperature 85 °C: —, 130 °C: - · - ·, WM: whole milk with 3.5% fat, SM: skim milk; according to *Kessler, 1988*)

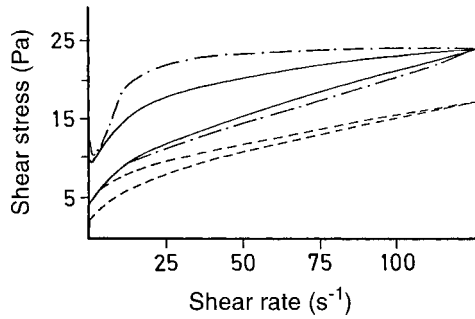


Fig. 10.11. Flow curves of stiff skim-milk yoghurt subjected to defined prestirring as a function of the rate of denaturation of β -lactoglobulin B (temperature/time/denaturation rate 90 °C/2.2 s/10%: ---, 90 °C/21 s/60%: —, 90 °C/360 s/99%: - · - ·; according to *Kessler, 1988*)

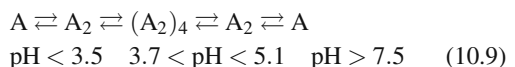
Flow curves of skim-milk yoghurt as a function of the rate of denaturation of β -lactoglobulin are presented in Fig. 10.11. The yield point is a measure of the elastic properties of the gel and the area enclosed by the hysteresis loop is a measure of the total energy required to destroy the gel. Both parameters increase with increasing rate of denaturation, which is a sign of increasing gel stability. In contrast to yoghurt production, syneresis of the gel is desirable in the production of cottage cheese, so that the typical texture is attained. For this reason, the milk is only slightly heat treated and the surface hydrophobicity is increased by the addition of chymosin before acidification.

10.1.2.1.4 Whey Proteins

β -Lactoglobulin occurs in genetic variants A, B and C of the Jersey dairy cattle breed, and variant D of the Montbeliarde dairy cow. Two other A_{Dr} and B_{Dr} variants of Australian drought master cows are identical to variants A and B apart from the carbohydrate content.

Table 10.9 shows the corresponding changes in the amino acid composition of β -lactoglobulin.

The monomeric β -lactoglobulin has a molecular weight of 18 kdal and consists of 162 amino acids, whose sequence is shown in Table 10.8. It exhibits a reversible, pH-dependent oligomerization, as represented by the equation:



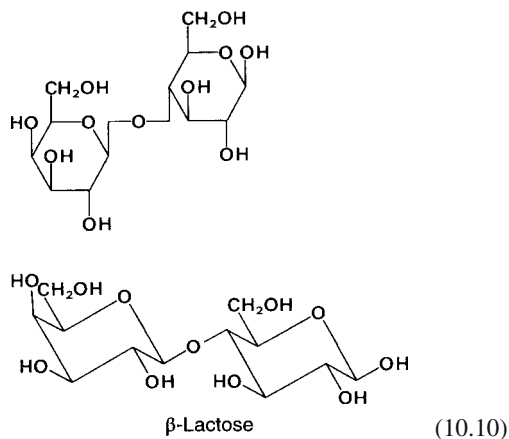
Hence, the monomer is stable only at a pH less than 3.5 or above 7.5. The octamer occurs with variant A, but not with variants B and C. Irreversible denaturation occurs at a pH above 8.6 as well as by heating or at higher levels of Ca^{2+} ions. β -lactoglobulin has 5 cysteine residues, one (Cys^{121} , Table 10.8) of which being free. In the native protein, however, this cysteine is buried within the structure. This SH group is exposed on partial denaturation and can participate either in protein dimerization via disulfide bridge formation or in reactions with other milk proteins, especially with κ -casein and α -lactalbumin, which proceed during the heating of milk.

α -Lactalbumin (M_r 14,200). This protein exists in two genetic forms, A and B ($\text{Gln} \rightarrow \text{Arg}$). It has 8 cysteine residues. Its amino acid sequence (Table 10.8), which is similar to that of lysozyme, has been elucidated. Disulfide bonds and a Ca^{2+} ion participate in the stabilization of the tertiary structure. α -Lactalbumin has a biological function since it is the B subunit of the enzyme lactose synthetase. The enzyme subunit A is a nonspecific UDP-galactosyl transferase; the subunit B makes sure that the transfer of the galactose residue can occur at the low glucose concentration present in mammals. The affinity of the transferase alone for glucose is too low ($K_m = 2 \text{ mol/l}$). It is increased 1000 fold by cooperation with α -lactalbumin.

10.1.2.2 Carbohydrates

The main sugar in milk is lactose, an O- β -D-galactopyranosyl-(1 $\rightarrow$ 4)-D-glucopyranose, which is 4–6% of milk.

The most stable form is α -lactose monohydrate, $\text{C}_{12}\text{H}_{22}\text{O}_{11} \cdot \text{H}_2\text{O}$. Lactose crystallizes in this form from a supersaturated aqueous solution at $T < 93.5^\circ\text{C}$. The crystals may have a prism- or pyramid-like form, depending on conditions. Vacuum drying at $T > 100^\circ\text{C}$ yields a hygroscopic α -anhydride. Crystallization from aqueous solutions above 93.5°C provides water-free β -lactose (β -anhydride, cf. Formula 10.10). Rapid drying of a lactose solution, as in milk powder production, gives a hygroscopic and amorphous equilibrium mixture of α - and β -lactose.



Some physical data of lactose are summarized in Table 10.15. The ratio of anomers is temperature dependent. As temperature increases, the β -form decreases. The mutarotation rate is temperature ($Q_{10} = 2.8$) and pH dependent (Fig. 10.12). The rise in mutarotation rate at $\text{pH} < 2$ and $\text{pH} > 7$ originates from the rate-determining step of ring opening, which is catalyzed by both H^+ and OH^- ions:

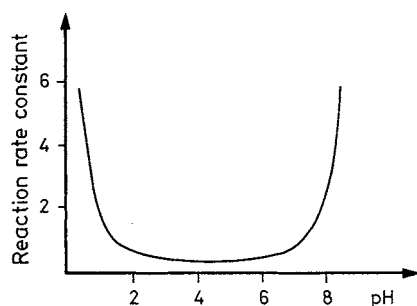
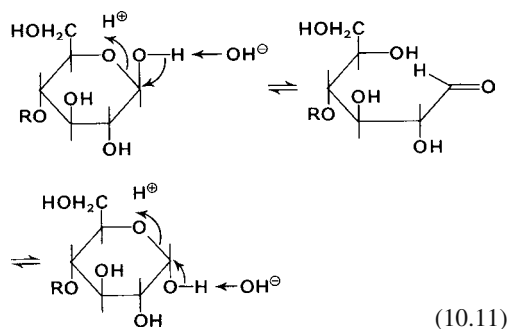


Fig. 10.12. Mutarotation rate of lactose as affected by pH

Table 10.15. Some physical characteristics of lactose

	α -Lactose	β -Lactose	Equilibrium mixture
Melting point (°C)	223 ^a	252.2 ^a	
Spec. rotation $[\alpha]_{\text{D}}^{20}$	89.4	35.0	
Equilibrium in aqueous solution ^b			
0 °C	1.00	1.80	
20 °C	1.00	1.68	
50 °C	1.00	1.63	
Solubility in water ^c			
0 °C	5.0	45.1	11.9
25 °C	8.6		21.6
39 °C	12.6		31.5
100 °C	70	94.7	157.6

^a Anhydrous. ^b Relative concentration.^c g Lactose/100 g water.

The great solubility difference between the two anomers is noteworthy. The sweetness of lactose is significantly lower than that of fructose, glucose or sucrose (Table 10.16). For people who suffer under lactose intolerance, dietetic milk products are produced by treatment with β -1,4-galactosidase (cf. 2.7.2.2.7). Glucose and some other amino sugars and oligosaccharides are present in small amounts in milk.

Lactulose is found in heated milk products. It is a little sweeter and clearly more soluble than lactose. For example, condensed milk contains up to 1% of lactulose, corresponding to an isomerization of ca. 10% of the lactose present. The formation proceeds via the *Lobry de Bruyn–van Ekenstein* rearrangement (cf. 4.2.4.3.2) or via *Schiff* base. Traces of epilactose (4-O- β -D-glacto-

Table 10.16. Relative sweetness of saccharose, glucose, fructose and lactose^a

Saccharose	Glucose	Fructose	Lactose
0.5	0.9	0.4	1.9
5.0	8.3	4.2	15.7
10.0	12.7	8.7	20.7
20.0	21.8	16.7	33.3

^a Results are expressed as concentration % for isosweet aqueous sugar solutions.

pyranosyl-D-mannose) are also formed on heating milk.

10.1.2.3 Lipids

The composition of milk fat is presented in Table 10.17. Milk fat contains 95–96% triglycerols. Its fatty acid composition is given in Table 10.18. The relatively high content of low molecular weight fatty acids, primarily of butyric acid, is characteristic of milk. Although linoleic acid dominates in the lipids occurring in feed, the content of this fatty acid is very low in milk fat (Table 10.17). The reason was found to be that microorganisms living in the rumen hydrogenate the linoleic acid to oleic acid and stearic acid with the formation of conjugated linoleic acid (CLA, cf. 3.2.1.2) and vaccenic acid as intermediates, as shown in Fig. 10.13. It is possible to increase the concentration of linoleic

Table 10.17. Milk lipids

Lipid fraction	Percent of the total lipid
Triacylglycerols	95–96
Diacylglycerols	1.3–1.6
Monoacylglycerols	0.02–0.04
Keto acid glycerides	0.9–1.3
Hydroxy acid glycerides	0.6–0.8
Free fatty acids	0.1–0.4
Phospholipids	0.8–1.0
Sphingolipids	0.06
Sterols	0.2–0.4

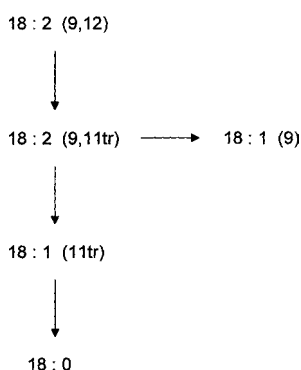
**Fig. 10.13.** Biohydrogenation of linoleic acid in the rumen of ruminants

Table 10.18. Fatty acid composition of milk fat^a

Fatty acid	Weight-%
Saturated, straight chain	
Butyric acid	2.79
Caproic acid	2.34
Caprylic acid	1.06
Capric acid	3.04
Lauric acid	2.87
Myristic acid	8.94
Pentadecanoic acid	0.79
Palmitic acid	23.8
Heptadecanoic acid	0.70
Stearic acid	13.2
Nonadecanoic acid	0.27
Arachidic acid	0.28
Behenic acid	0.11
Saturated, branched chain	
12-Methyltetradecanoic acid	0.23
13-Methyltetradecanoic acid	0.14
14-Methylpentadecanoic acid	0.20
14-Methylhexadecanoic acid	0.23
15-Methylhexadecanoic acid	0.36
3,7,11,15-Tetramethylhexadecanoic acid	0.12–0.18
Unsaturated	
9-Decenoic acid	0.27
9-cis-Tetradecenoic acid	0.72
9-cis-Hexadecenoic acid	1.46
9-cis-Heptadecenoic acid	0.19
8-cis-Octadecenoic acid	0.45
Oleic acid	25.5
11-cis-Octadecenoic acid	0.67
9-trans-Octadecenoic acid	0.31
10-trans-Octadecenoic acid	0.32
11-trans-Octadecenoic acid	1.08
12-trans-Octadecenoic acid	0.12
13-trans-Octadecenoic acid	0.32
14-trans-Octadecenoic acid	0.27
15-trans-Octadecenoic acid	0.21
16-trans-Octadecenoic acid	0.23
Linoleic acid	2.11
Linolenic acid	0.38

^a Only acids with a content higher than 0.1% are listed.

acid in milk fat, e.g., by adding plant fats of the appropriate composition in encapsulated form to the feed. The disadvantage of such a nutritionally/physiologically interesting approach is

the changed physico-chemical properties of the dairy product, e.g., an increased susceptibility to oxidation and the formation of unsaturated lactones (γ -dodec-cis-6-enolactone from linoleic acid) which influences the flavor of milk and meat. In addition to the main straight-chain fatty acids, small amounts of odd-C-number, branched-chain and oxo-fatty acids (cf. 3.2.1.3) are present.

Phospholipids are 0.8–1.0% of milk fat and sterols, mostly cholesterol, are 0.2–0.4%. Butterfat melting properties, as affected by season and fodder, are listed in Table 10.19. Milk fat is very finely distributed in plasma. The diameter of fat globules is 0.1–10 μm , but for the main part in the range of 1–5 μm . During homogenization, milk at 50–75 °C is forced through small passages under pressure of up to 35 MPa, the diameter of the globules lowers to <1 μm , depending on the pressure. The fat droplets are surrounded by a membrane that consists of phospholipids and a double layer of proteins and accounts for about 2% of the total mass of the droplet. The layer thickness is on average 8–9 nm, but is not uniform. Membrane compositional data are given in Table 10.20.

About 40 proteins with M_r 15–240 kdal (milk fat globule membrane proteins, MFGM proteins) are involved in the make-up of the membrane. Although their nutritional value with regard to the caseins and whey proteins is low, they receive

Table 10.19. Melting characteristics of butterfat

Temperature (°C)	Solid content (%)	Temperature (°C)	Solid content (%)
5	43–47	30	6–8
10	40–43	35	1–2
20	21–22	40	0

Table 10.20. Membrane composition of milkfat globules

Constituent	Proportion (%)
Protein	41
Phospho- and glycolipids	30
Cholesterol	2
Neutral glycerides	14
Water	13

attention because they can be detrimental to the health of sensitized persons. Casein proteins enter and participate in membrane formation when the fat globule surface area is expanded 4- to 6-fold during homogenization of milk. Six of the eight dominant proteins in the MFGM are glycoproteins, e.g., xanthine oxidoreductase. Other enzymes which are present in the membrane are acetylcholine esterase as well as alkaline and acidic phosphatase (cf. Table 10.24). A very active lipoprotein lipase, a glycoprotein (8.3% carbohydrate, molecular weight 48.3 kdal), occurs in the casein micelles. However, if the milking and storage procedures are appropriate, the raw milk can be kept for several days without the development of a rancid off-flavor. It is likely that the membranes of the fat globules prevent lipolysis. Disruption of the organized structure of the membrane, for instance by homogenization, allows the lipase to bind to the fat globules and to hydrolyze the triacylglycerols at a high rate (1 μ mole fatty acid per min per ml milk, pH 7, 37 °C). The milk becomes unpalatable within a few minutes. Therefore the lipoprotein lipase has to be inactivated by pasteurization prior to milk homogenization.

The small amounts of gangliosides that occur in milk (5.6 μ mol/l, calculated as ganglioside-bound sialic acid) are of interest for the analytical differentiation of skim-milk and butter-milk powder. As structural elements of the membrane of the fat globules, the cream gets enriched with gangliosides during skimming and only about 8% remain in the skimmed milk. During butter-making, the membrane of the fat globules is mechanically destroyed and the highly polar gangliosides pass almost completely into the buttermilk. Therefore, unlike skim-milk powder, butter-milk powder is rich in gangliosides (ca. 480 μ mol/kg, calculated as sialic acid).

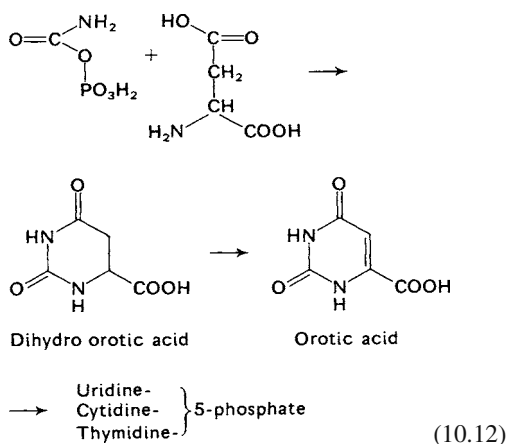
10.1.2.4 Organic Acids

Citric acid (1.8 g/l) is the predominant organic acid in milk. During storage it disappears rapidly as a result of the action of bacteria. Other acids (lactic, acetic) are degradation products of lactose. The occurrence of orotic acid (73 mg/l), an intermediary product in biosynthesis of pyrimidine nucleotides, is specific for milk:

Table 10.21. Indicators for the proportion of milk in foods

Compound	Whole milk powder	Skim milk powder
Orotic acid		
photometric	50.6	66.4
polarographic	46.6	58.1
Total creatinine	66.3	84.4
Uric acid	12.4	15.3

Expressed as mg/100 g solids.



Orotic acid as well as total creatinine and uric acid are suitable indicators for the determination of the proportion of milk in foods. The average values for whole-milk and skim-milk powder given in Table 10.21 can serve as reference values.

10.1.2.5 Minerals

Minerals, including trace elements, in milk are compiled in Table 10.22.

10.1.2.6 Vitamins

Milk contains all the vitamins in variable amounts (Table 10.23). During processing, the fat-soluble vitamins are retained by the cream, while the water-soluble vitamins remain in skim milk or whey.

Table 10.22. Mineral composition of milk

Constituent	mg/l	Constituent	µg/l
Potassium	1500	Zinc	4000
Calcium	1200	Aluminum	500
Sodium	500	Iron	400
Magnesium	120	Copper	120
Phosphate	3000	Molybdenum	60
Chloride	1000	Manganese	30
Sulfate	100	Nickel	25
		Silicon	1500
		Bromine	1000
		Boron	200
		Fluorine	150
		Iodine	60

Table 10.23. Vitamin content of milk

Vitamin	mg/l	Vitamin	mg/l
A (Retinol)	0.4	Nicotinamide	1
D (Calciferol)	0.001	Pantothenic acid	3.5
E (Tocopherol)	1.0	C (Ascorbic acid)	20
B ₁ (Thiamine)	0.4	Biotin	0.03
B ₂ (Riboflavin)	1.7	Folic acid	0.05
B ₆ (Pyridoxine)	0.6		
B ₁₂ (Cyanocobalamine)	0.005		

10.1.2.7 Enzymes

Milk contains a great number of enzymes which are not only of analytical importance for the detection of heat-treated milk, but can also influence the processing properties. The rate of heat inactivation of the enzymes indicates the type and extent of heating (cf. 2.5.4).

Hydrolases identified include: amylases, lipases, esterases, proteinases and phosphatases. Proteinase inhibitors have also been found. Important oxidoreductases in milk are aldehyde dehydrogenase (xanthine oxidase), lactoperoxidase and catalase. A general idea of the occurrence and localization of enzymes in bovine milk is given in Table 10.24

10.1.2.7.1 Plasmin

The serine endopeptidase plasmin is of special interest in milk technology.

Table 10.24. Enzymes in bovine milk

EC Number	Name	Localization ^a
1.1.1.27	L-Lactate dehydrogenase	P
1.1.1.37	Malate dehydrogenase	
1.1.3.22	Xanthine oxidase	F
1.4.3.6	Amine oxidase (copper-containing)	
1.6.99.3	NADH dehydrogenase	F
1.8	Sulfhydryl oxidase ^b	S
1.8.1.4	Dihydrolipoamide dehydrogenase	F
1.11.1.6	Catalase	L
1.11.1.7	Lactoperoxidase	S
1.15.1.1	Superoxide dismutase	
2.3.2.2	γ-Glutamyltransferase	F
2.4.1.22	Lactose synthase	S
2.4.99.1	β-Galactoside-α-2,6-sialyltransferase	
2.6.1.1	Aspartate aminotransferase	P
2.6.1.2	Alanine aminotransferase	
2.7.1.26	Riboflavin kinase	
2.7.1.30	Glycerol kinase	
2.7.7.2	FMN adenyl transferase	
2.8.1.1	Thiosulfate sulfurtransferase	
3.1.1.1	Carboxylesterase	S
3.1.1.2	Arylesterase	
3.1.1.7	Acetylcholine esterase	F
3.1.1.8	Choline esterase	S
3.1.1.34	Lipoproteinlipase	C
3.1.3.1	Alkaline phosphatase	F
3.1.3.2	Acid phosphatase	F
3.1.3.5	5'-Nucleotidase	F
3.1.3.9	Glucose-6-phosphatase	F
3.1.3.16	Phosphoprotein phosphatase	P
3.1.4.1	Phosphodiesterase I	F
3.1.27.5	Pancreatic ribonuclease	S
3.2.1.1	α-Amylase	S
3.2.1.2	β-Amylase	
3.2.1.17	Lysozyme	S
3.2.1.24	α-Mannosidase	
3.2.1.31	β-Glucuronidase	
3.2.1.52	β-N-Acetylglucosaminidase	
3.4	Acid peptidases	
3.4.21.7	Plasmin	C
3.6.1.3	Adenosinetriphosphatase	F
3.6.1.9	Nucleotide pyrophosphatase	
4.1.2.13	Fructosebiphosphate aldolase	
5.3.1.9	Glucose-6-phosphate isomerase	

^a C: casein micelle, F: fat-globule membrane, L: leucocytes, P: plasma, S: serum.

^b Not thiol oxidase EC 1.8.3.2.

Plasmin, its precursor plasminogen and the plasminogen activator (PA) are present in milk associated to the casein micelles and the membranes of the fat globules. A shift of the pH value to the acidic range (pH 4.7) promotes the release of plasmin from casein. The concentration of plasmin is 0.3–2.5 µg/ml milk. It increases with the age of the cow and during the lactation period. Plasmin is in the pH range 7.5–8.0 most active at 37 °C. It preferentially hydrolyzes β-casein and at a lower rate also α-casein. κ-Casein is resistant just like the whey proteins α-lactalbumin and β-lactoglobulin. Like trypsin, it attacks the carboxyl side of L-lysine as well as L-arginine on hydrolysis. The activity of plasmin is controlled by the PA inhibitor and the plasmin inhibitor.

The plasmin activity is reduced by only 10–17% under the conditions of pasteurization, e. g., 72 °C for 15 s. Storage of pasteurized milk indirectly promotes plasmin activity because the inhibitors of PA are inactivated. Complete thermal inactivation of plasmin is achieved at 120 °C in 15 min and at 142 °C in 18 s.

Plasmin influences the ripening process, e. g., in Camembert. It is accelerated and aroma formation is improved. In the recovery of caseinates, on the other hand, the separation of plasmin is absolutely necessary.

10.1.2.7.2 Lactoperoxidase

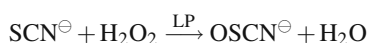
In the preservation of raw milk, the antimicrobial lactoperoxidase (LP) system present in milk is of interest. This LP system offers an alternative, especially in countries where it is not possible to cool the milk to protect it from spoilage.

The system consists of LP (EC 1.11.1.7) and the substrates thiocyanate and H₂O₂. The enzyme, a glycoprotein (carbohydrate content 10%), consists of 612 amino acid residues (M_r 78,000, IP 9.6) and Fe-protoporphyrin IX, which as the prosthetic group carries out the catalysis, as described in 2.3.2.2. Thiocyanate takes part in this reaction process as the electron donor AH. Lactoperoxidase is one of the heat stable enzymes of milk, especially when the structure is fixed by Ca²⁺ ions. It is still active after 30 min at 63 °C (neutral pH) and after 15 s at 72 °C, but it is inactive after only 2.5 s at 80 °C. Acidification (pH 5.3) accelerates the inactivation by liberating the Ca²⁺ ions. After

xanthine oxidase, LP is the most common enzyme in milk: ca. 30 mg/l.

The thiocyanate concentration in milk depends on the feed, e. g., on the occurrence of glucosinolates (cf. 17.1.2.9.3). H₂O₂ is not a component of milk, but is supplied by bacteria, e. g., lactic acid bacteria.

Hypothiocyanite is the main product formed from the hydrogen donor thiocyanate in LP catalysis.



This compound is a bactericidal agent because it can oxidize the SH groups of structure-forming proteins and enzymes in bacteria. The LP system is used to prolong the shelf life of raw milk and pasteurized milk. H₂O₂ is produced here by the addition of glucose/glucose oxidase (cf. 2.7.2.1.1) and the concentration of thiocyanate is increased by addition.

In the production of fermented milk products, the LP system can inhibit the development of starter cultures.

10.1.3 Processing of Milk

Only a small amount of milk is sold to the consumer without processing (certified raw milk). The main part is subjected to a processing procedure shown schematically in Fig. 10.14.

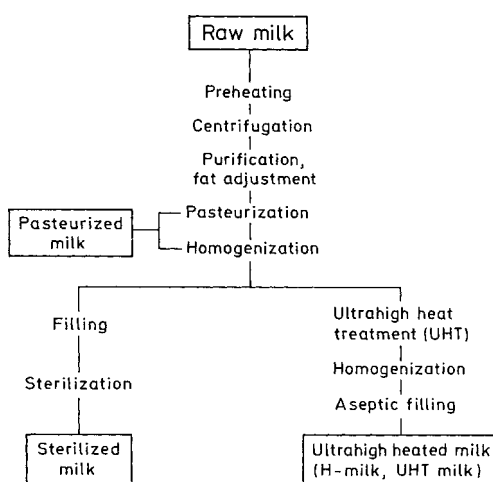


Fig. 10.14. Treatment of milk

10.1.3.1 Purification

The milk is usually delivered in the cooled tank (at least -8°C of a milk truck. For purification, it is fed into a clarifier (self-cleaning disk separator) via a deaerating vessel. These separators can process either cold or warm milk (40°C) at speeds of 4500–8400 rpm with throughput capacities of up to 50,000 l/h.

10.1.3.2 Creaming

After heating to about 40°C (increase in creaming efficiency by lowering the viscosity), the milk is separated into cream and skimmed milk in a cream separator. Cream separators have a nominal capacity of up to 25,000 l/h at speeds of 4700–6500 rpm. The fat content of the milk can be standardized by careful back-mixing.

10.1.3.3 Heat Treatment

The fluid milk is heated to improve its durability and to kill disease-causing microorganisms. Heat treatments used are (cf. Fig. 10.15):

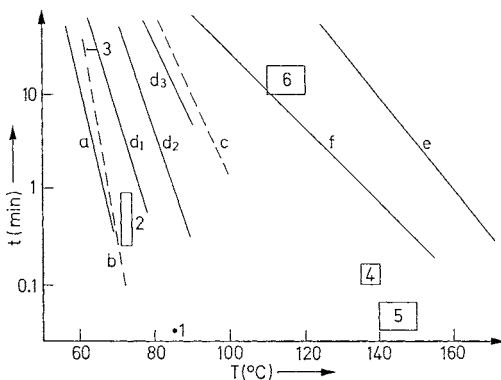


Fig. 10.15. Heating of milk. 1–3 Pasteurization: 1 high temperature treatment, 2 short time and 3 long time heat treatment; 4 and 5 UHT treatment: 4 indirect and 5 direct; 6 sterilization. *a*: Killing pathogenic microorganisms (*Tubercle bacilli* as labelling organism), *b/c*: inactivation of alkaline/acid phosphatase. *d*₁, *d*₂, *d*₃ denaturation (5, 40, 100%) of whey proteins. *e*: casein heat coagulation, *f*: start of milk browning

• Thermization.

The process involves heating under conditions that are milder than those of pasteurization, e. g., 57 – 68°C . The number of bacteria is reduced, e. g., for the production of cheese. The taste of the milk and the coagulation time during treatment with rennet are not impaired.

• Pasteurization.

The milk is treated: at high temperature (85°C for 2–3 s) in a short-time, flash process (72 – $75^{\circ}\text{C}/15$ – 30 s) in plate heaters; or by the low temperature or holder process, in which it is heated at 63 – 66°C for at least 30–32 min, with stirring, and is then cooled.

• Ultrahigh Temperature (UHT) Treatment.

The process involves indirect heating by coils or plates at 136 – 138°C for 5–8 s, or direct heating by live steam injection at 140 – 145°C for 2–4 s, followed by aseptic packaging.

To prevent dilution or concentration of the milk, the amount of injected steam must be controlled in such a way that it corresponds to the amount of water withdrawn during expansion under vacuum.

• Bactotherm Process.

This is a combination of centrifugal sterilization in bactofuges (65 to 70°C) and UHT heating of the separated sediment (2–3% of the milk), followed by recombination. Since the total amount of milk is not heated in this process, the taste is improved. The storability is ca. 8–10 days.

• Sterilization.

Milk in retail packages is heated in autoclaves at 107 – $115^{\circ}\text{C}/20$ – 40 min, 120 – $130^{\circ}\text{C}/8$ – 12 min.

10.1.3.4 Homogenization

Homogenization is conducted to stabilize the emulsion milk by reducing the size of the fat globules. This is achieved by high-pressure homogenization (up to 35 MPa, 50 – 75°C). In principle, the high-pressure homogenizer is a high-pressure pump which presses the product through a homogenizing valve. The fat globules are reduced in size to a diameter of $<1\mu\text{m}$ by turbulence, cavitation and shear forces, resulting in a ca. 10 fold increase in the surface area. The membranes of the reduced fat globules

are formed by the uptake of caseins and whey proteins.

10.1.3.5 Reactions During Heating

Heat treatment affects several milk constituents. Casein, strictly speaking, is not a heat-coagulable protein; it coagulates only at very high temperatures (cf. Fig. 10.15). Heating at 120 °C for 5 h dephosphorylates sodium or calcium caseinate solutions (100% and 85%, respectively) and releases 15% of the nitrogen in the form of low molecular weight fragments.

However, temperature and pH strongly affect casein association and cause changes in micellar structure (cf. 10.1.2.1.2 and 10.1.2.1.3). An example of such a change is the pH-dependent heat coagulation of skim milk. The coagulation temperature drops with decreasing pH (Fig. 10.16 and 10.9). Salt concentration also has an influence, e.g., the heat stability of milk decreases with a rise in the content of free calcium.

All pasteurization processes supposedly kill the pathogenic microorganisms in milk. The inactivation of the alkaline phosphatase is used in determining the effectiveness of pasteurization. At higher temperatures or with longer heating times, the whey proteins start to denature – this coincides with the complete inactivation of acid phosphatase. Denatured whey proteins, within the pH-range of their isoelectric points, cease to be soluble and coagulate together with casein due to souring or chymosin action of the milk. Such co-precipitation of the milk proteins is of importance

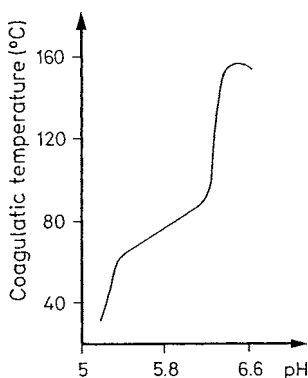


Fig. 10.16. Thermal coagulation of skim milk

in some milk processing (as in cottage cheese production). The thermal stability of whey proteins is illustrated in Fig. 10.17.

Heat treatment of milk activates thiol groups; e.g., a thiol-disulfide exchange reaction occurs between κ -casein and β -lactoglobulin. This reaction reduces the vulnerability of κ -casein to chymosin, resulting in a more or less strong retardation of the rennet coagulation of heated milk.

Further changes induced by heating of milk are:

- Calcium phosphate precipitation on casein micelles.
- *Maillard* reactions between lactose and amino groups (e.g. lysine) which, in a classical sterilization process, causes browning of milk and formation of hydroxymethyl furfural (HMF).
- δ -Lactone and methyl ketone formation from glycerides esterified with hydroxy- or keto-fatty acids.
- Degradation of vitamin B₁, B₆, B₁₂, folic acid and vitamin C. Losses of 10–30% in the production of UHT milk are possible. Sterilization destroys ca. 50% of the vitamins B₁, B₆ and folic acid and up to 100% of vitamin C and B₁₂.

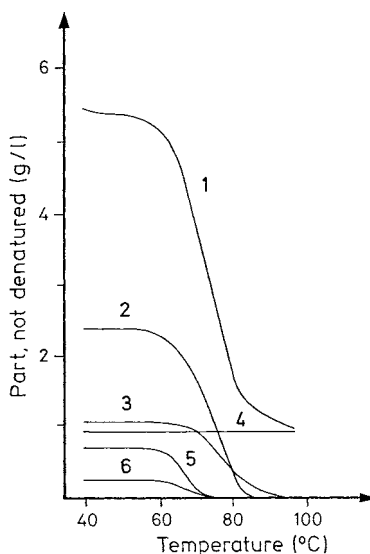


Fig. 10.17. Denaturation of whey proteins by heating at various temperatures for 30 min. 1 Total whey protein, 2 β -lactoglobulin, 3 α -lactalbumin, 4 proteose peptone, 5 immunoglobulin, 6 serum albumin

- Changes in membranes of milk fat globules, which affect the cream separation property of the globules.

Detailed studies have shown that the rate of several reactions which occur during heating of milk, e.g., thiamine and lysine degradations, formation of HMF and nonenzymic browning, can be calculated over a great temperature-time range (including extended storage) by application of a second-order rate law. Assuming an average activation energy of $E_a = 102$ kJ/mole, a “chemical effect” $C^* = 1$ has been defined which gives a straight line in a $\log t$ vs. T^{-1} diagram, from which the thiamine loss is seen to be approx. 0.8 mg/l (Fig. 10.18). Other lines in Fig. 10.18 represent a power of ten of heat treatment and chemical reactions ($C^* = 10^{-1}, 10^{-2}, \dots$, or $10^1, 10^2, \dots$). The pigments formed in a browning reaction become visible only in the range of $C^* = 10$.

Quality deterioration in the form of nutritional degradation, changes in color or development of off-flavor have also been predicted for other foods by application of a suitable mathematical model.

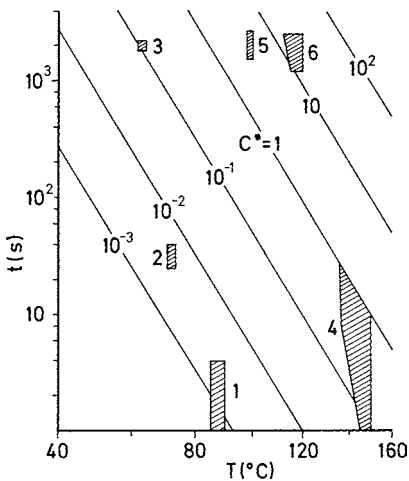


Fig. 10.18. Chemical reactions in heat-treated milk. (“Chemical effect” $C^* = 1$: losses of approx. 3% thiamine and approx. 0.7% lysine and formation of approx. 0.8 mg/l HMF); commonly used heat treatments: 1 high heat, 2 short time heating, 3 prolonged heating, 4 UHT treatment, 5 boiling, 6 sterilization (according to Kessler, 1983).

HMF: Hydroxymethylfurfural

In most cases the loss of quality fits a zero- or first-order rate law. Knowledge of the rate constant allows one to predict the extent of reaction for any time.

The influence of temperature on the reaction rate follows the *Arrhenius* equation (cf. 2.5.4). Thus by studying a reaction and measuring the rate constants at two or three high temperatures, one could then extrapolate with a straight line to a lower temperature and predict the rate of the reaction at the desired lower temperature. However, these data allow only a prediction of the shelf life when the physical and chemical properties of the components of a food do not alter with temperature. For example, as temperature rises a solid fat goes into a liquid state. The reactants may be mobile in the liquid fat and not in the solid phase. Thus, shelf life will be underestimated for the lower temperature.

10.1.4 Types of Milk

Milk is consumed in the following forms:

- *Raw fluid milk* (high quality milk), which has to comply with strict hygienic demands.
- *Whole milk* is heat-treated and contains at least 3% fat. It can be a standardized whole milk adjusted to a predetermined fat content, in which case the fat content has to be at least 3.5%.
- *Low-fat milk* is heat-treated and the cream is separated. The fat content is 1.5–2%.
- *Skim milk* is heat-treated and the fat content is less than 0.3%.
- *Reconstituted milk* is most common in regions where milk production is not feasible (e.g. many Japanese cities). For production, melted butter fat is emulsified in a suspension of skim milk powder at 45 °C. The “cream” with a fat content of 20–30% is subjected to two-stage homogenization (20 and 5 MPa, 55–60 °C) and then diluted with the skim milk suspension.
- *Filled milk* is less expensive because the butter fat is replaced with a plant fat.
- *Toned milk* is a blend of a fat-rich fresh milk and reconstituted skim milk in which the non-fat solids are “toned up”. Addition of water “tones down” the fat and nonfat solids.

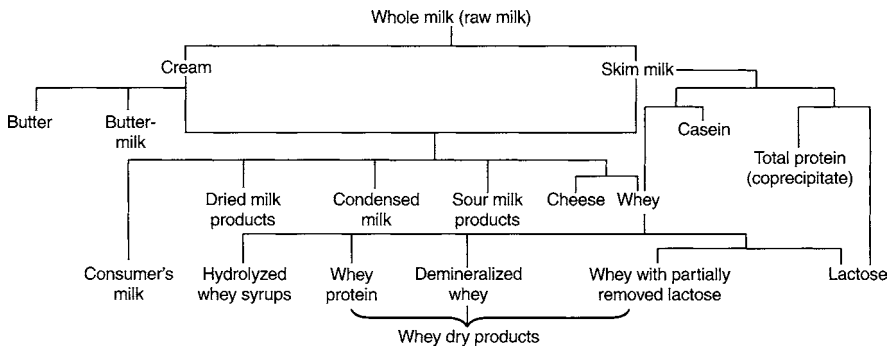


Fig. 10.19. Schematic presentation of milk processing